Wealth in the Phillips Curve

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- **The question.** Inflation has remained above the Federal Reserve's 2 percent target for more than five years, even as wage growth has slowed and the labor market is not a source of inflationary pressure. This raises the question of whether labor-market slack adequately captures demand pressure, or whether accumulated wealth helps explain persistent excess inflation.
- **Time-series evidence.** Adding combined stock and housing wealth to an otherwise standard Phillips curve raises R^2 from 0.063 to 0.248; including income and activity controls raises it to 0.519. The model with these controls fits 0.86 percentage point of excess inflation in 2026Q2, approximately nine-tenths of the observed gap of 0.95 percentage point, compared with essentially zero in the slack-only model (Table 1; Figures 1–2).
- **State-level evidence.** Time-series evidence is subject to confounding factors. We therefore complement the aggregate analysis with state-level evidence, following Hazell et al. (2022). State and year fixed effects absorb persistent geographic differences and common national shocks. Stock wealth uses dividend capitalization motivated by Chodorow-Reich et al. (2021). Housing wealth revalues the fixed Census 2000 owner-occupied stock with state house-price indexes, motivated by the housing-wealth channel in Stroebel and Vavra (2019).
- **The magnitude.** Combined wealth is positive and marginally significant without income controls ($p = 0.062$), and significant with dividend and interest changes ($p = 0.004$; Table 2). With controls, a 10 percentage point higher wealth-to-income ratio is associated with 0.08 percentage point higher nontradeable inflation. Applying the combined coefficient to national net worth above its 2005–2017 mean gives an illustrative 1.7 percentage points, conditional on transferring the local coefficient to aggregate net worth.
- **The mechanism.** In a traditional New Keynesian Phillips curve, the inflationary effects of wealth should already be captured by marginal costs and expected inflation. However, wealth may affect inflation beyond labor-market slack by changing firms' desired markups. Wealthier households may become less price-sensitive and less willing to switch retailers when prices rise, allowing firms to charge higher markups even without higher production costs (Stroebel and Vavra, 2019; Mongey and Waugh, 2025). This is a possible interpretation of the estimates.

Table 1: Household resources in the national Phillips curve

$$\pi_t - e_t = \alpha + \beta(u_t - u_t^*) + \boldsymbol{\theta}'\mathbf{W}_t + \gamma' X_t + \varepsilon_t. \tag{1}$$

Dependent variable:	Excess core PCE inflation			
	(1)	(2)	(3)	(4)
Unemployment gap	-0.127^{*}	-0.030	-0.064	-0.055
	(0.067)	(0.056)	(0.076)	(0.075)
Wealth		0.613^{***}	0.763^{***}	
		(0.212)	(0.192)	
Stock wealth				0.618^{***}
				(0.231)
Housing wealth				0.984^{***}
				(0.367)
Controls			✓	✓
R^2	0.063	0.248	0.519	0.524
Observations	187	187	187	187

Notes: Sample: 1979Q4–2026Q2. $\pi_t - e_t$ is four-quarter core PCE log inflation less long-run expected inflation, in percentage points. Wealth is household/nonprofit directly held corporate equities (FRED HNOCEA) plus real estate (HNOREMV), divided by annualized DPI_{t-4}. Column (1) includes only the unemployment gap and an intercept. Columns (2)–(3) add combined wealth; (4) separates the components. This two-asset measure excludes other net wealth and parallels the state table; the national and state measurement methods differ. Controls in (3)–(4) (X_t) comprise asset-income growth (interest, dividends and rent), the output gap, and consumption growth. All columns include an intercept. Parentheses: four-lag Newey–West HAC standard errors; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Expectations splice Blue Chip/Livingston and SPF with the project's CPI-to-PCE adjustment. Estimates are full-sample conditional associations. The slack-only benchmark has $R^2 = 0.063$. Specification choice followed exploratory checks; stars use nominal p-values. Stock wealth in (4) reverses sign through 2019 (-0.311, $p = 0.060$); its full-sample $p = 0.007$ becomes 0.222 after Holm adjustment across 35 checks. *Sources:* BEA; Financial Accounts/FRED; project labor-market and survey inputs.

National and state evidence

Figure 1: Excess inflation and fitted values

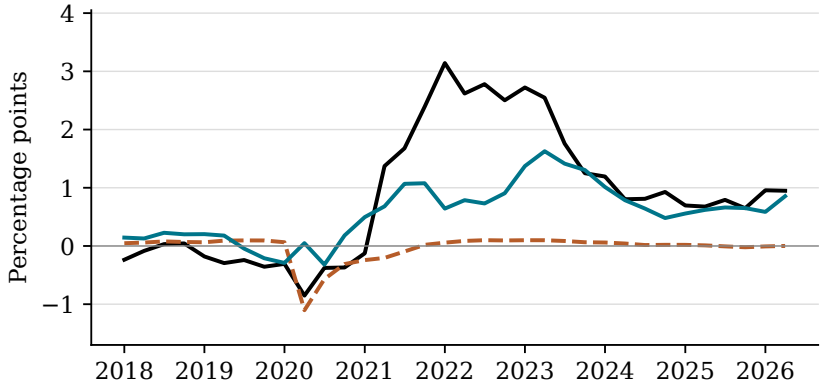
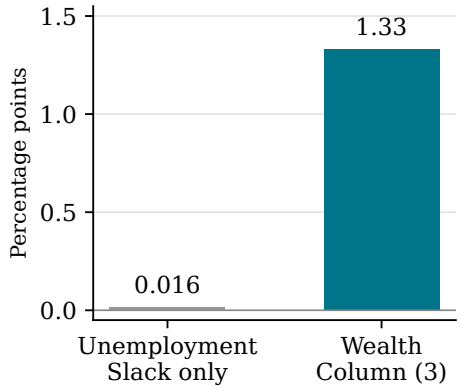


Figure 2: 2026Q2 contributions



Observed Slack only Column (3)

Notes: Figure 1: observed excess inflation is 0.95 percentage point in 2026Q2; the slack-only fit is 0.00 and Table 1, column (3), fits 0.86. Figure 2: unemployment uses the slack-only model, relative to a zero gap; wealth uses column (3), relative to sample-mean wealth, $0.763(4.636 - 2.890) \simeq 1.33$. The bars come from different specifications and do not add to fitted inflation. These are full-sample fits, not out-of-sample forecasts. *Source:* Author's calculations.

Table 2: State Phillips curves and household wealth

$$\pi_{it}^{N} = \alpha_i + \lambda_t + \beta \bar{u}_{it}^{(4)} + \boldsymbol{\theta}' \mathbf{W}_{i,t-1} + \rho \bar{p}_{it}^{N,(4)} + \gamma' X_{it} + \varepsilon_{it}. \tag{2}$$

Dependent variable:	Nontradeable inflation			
	(1)	(2)	(3)	(4)
Unemployment	0.062	0.146	0.196	0.189
	(0.094)	(0.118)	(0.119)	(0.121)
Wealth		0.478*	0.822***	
		(0.248)	(0.269)	
Stock wealth				0.956***
				(0.346)
Housing wealth				0.507
				(0.411)
Controls			✓	✓
Within R^2	0.0577	0.0673	0.0991	0.1004
Observations	442	442	442	442

Notes: Identical annual sample, 2005–2017, 33 states and DC. Outcome: annual mean of quarterly year-on-year nontradeable inflation excluding shelter. $\bar{u}^{(4)}$ and $\bar{p}^{N,(4)}$ are annual means of four-quarter-lagged unemployment and relative prices. $\mathbf{W}$ is absent in (1), combined stock and housing wealth in (2)–(3), and the two components separately in (4). All columns include unemployment, lagged relative prices, and state and year fixed effects. The Controls checkmark denotes the additional dividend and monetary-interest changes, scaled by DPI (X), included together in (3)–(4). Wealth is divided by same-year DPI and then lagged one year. Parentheses: state-clustered CR1 standard errors with t_{33} inference; stars as in Table 1. The illustrative scaling is $0.822(8.55 - 6.48) \simeq 1.70$; the local coefficient and wealth definition differ from their national counterparts. Housing wealth is a Census/FHFA fixed-stock proxy motivated by Stroebel and Vavra (2019), not their exact measure or identification design. *Sources:* HHNS; BEA; Shiller; Census 2000; FHFA.

References

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